

Data Analytics (DA)

Module -2

Power BI Assignment 2 – DAX & Data Visualization

Tasks:

DAX Calculations

1. Calculated Columns

- Create a Calculated Column for 'Category Type': Add a calculated column in the Order Details table that combines the 'Category' and 'Sub-Category' columns into a single 'Category Type' column.
- Calculate Revenue per Order in Order Details Table: Create a calculated column in the Order Details table to compute the revenue (Amount * Quantity) per order.
- Create a Calculated Column to Categorize Sales: Add a calculated column named 'Sales Category' in the Order Details table that categorizes each order as 'Above Average' or 'Below Average' based on the Amount value.

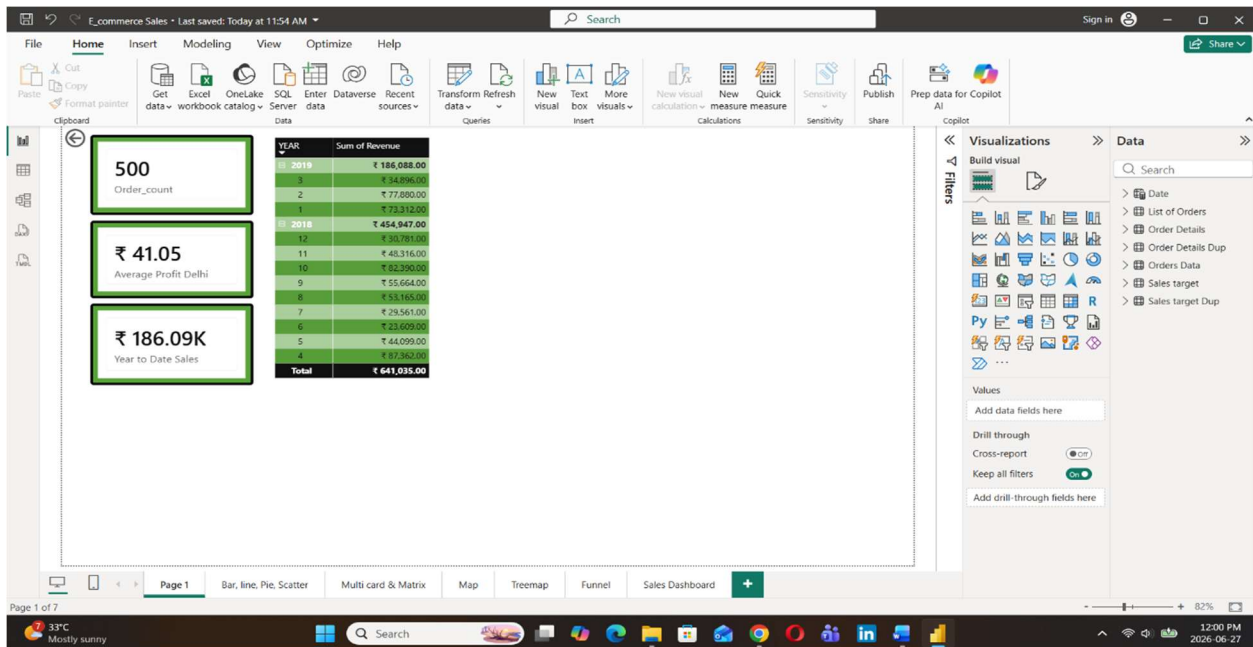
Order ID	Amount	Profit	Quantity	Category	Sub-Category	Profit Margin	Profit Status	Category Type	Revenue	Sales Category
B-25604	₹ 65.00	17	2	Clothing	T-Shirt	26.15%	Profit	Clothing, T-Shirt	₹ 130.00	Below Average
B-25606	₹ 87.00	4	2	Clothing	Shirt	4.60%	Profit	Clothing, Shirt	₹ 174.00	Below Average
B-25607	₹ 50.00	15	4	Clothing	Leggings	30.00%	Profit	Clothing, Leggings	₹ 200.00	Below Average
B-25615	₹ 68.00	20	5	Clothing	Hankchief	29.41%	Profit	Clothing, Hankchief	₹ 340.00	Below Average
B-25616	₹ 42.00	12	5	Clothing	Hankchief	28.57%	Profit	Clothing, Hankchief	₹ 210.00	Below Average
B-25619	₹ 353.00	90	8	Clothing	Saree	25.50%	Profit	Clothing, Saree	₹ 2,824.00	Above Average
B-25623	₹ 53.00	1	4	Clothing	Stole	1.89%	Profit	Clothing, Stole	₹ 212.00	Below Average
B-25624	₹ 26.00	12	3	Clothing	Hankchief	46.15%	Profit	Clothing, Hankchief	₹ 78.00	Below Average
B-25625	₹ 97.00	29	2	Clothing	Hankchief	29.90%	Profit	Clothing, Hankchief	₹ 194.00	Below Average
B-25628	₹ 45.00	13	4	Clothing	Skirt	28.89%	Profit	Clothing, Skirt	₹ 180.00	Below Average
B-25629	₹ 1,560.00	421	3	Clothing	Trousers	26.99%	Profit	Clothing, Trousers	₹ 4,680.00	Above Average
B-25630	₹ 133.00	12	5	Clothing	Stole	9.02%	Profit	Clothing, Stole	₹ 665.00	Below Average
B-25635	₹ 40.00	16	3	Clothing	Hankchief	40.00%	Profit	Clothing, Hankchief	₹ 120.00	Below Average
B-25636	₹ 637.00	113	5	Clothing	Saree	17.74%	Profit	Clothing, Saree	₹ 3,185.00	Above Average
B-25637	₹ 117.00	14	3	Clothing	Shirt	11.97%	Profit	Clothing, Shirt	₹ 351.00	Below Average
B-25642	₹ 434.00	26	11	Clothing	Shirt	5.99%	Profit	Clothing, Shirt	₹ 4,774.00	Above Average
B-25654	₹ 34.00	12	3	Clothing	Hankchief	35.29%	Profit	Clothing, Hankchief	₹ 102.00	Below Average
B-25658	₹ 27.00	9	2	Clothing	T-Shirt	33.33%	Profit	Clothing, T-Shirt	₹ 54.00	Below Average
B-25659	₹ 148.00	72	7	Clothing	T-Shirt	48.65%	Profit	Clothing, T-Shirt	₹ 1,036.00	Below Average
B-25670	₹ 74.00	29	3	Clothing	Stole	39.19%	Profit	Clothing, Stole	₹ 222.00	Below Average
B-25687	₹ 17.00	6	1	Clothing	Stole	35.29%	Profit	Clothing, Stole	₹ 17.00	Below Average
B-25690	₹ 31.00	10	3	Clothing	Skirt	32.26%	Profit	Clothing, Skirt	₹ 93.00	Below Average
B-25691	₹ 714.00	56	4	Clothing	Saree	7.84%	Profit	Clothing, Saree	₹ 2,856.00	Above Average
B-25694	₹ 167.00	43	7	Clothing	T-Shirt	25.75%	Profit	Clothing, T-Shirt	₹ 1,169.00	Below Average
B-25695	₹ 171.00	14	9	Clothing	Shirt	8.19%	Profit	Clothing, Shirt	₹ 1,539.00	Below Average
B-25717	₹ 561.00	212	3	Clothing	Saree	37.79%	Profit	Clothing, Saree	₹ 1,683.00	Above Average
B-25719	₹ 29.00	10	2	Clothing	Stole	34.48%	Profit	Clothing, Stole	₹ 58.00	Below Average
B-25746	₹ 87.00	16	7	Clothing	Stole	10.30%	Profit	Clothing, Stole	₹ 174.00	Below Average

2. Calculated Measures

1. **Order Count-** Create a measure that calculates the total number of orders in the Order Details table.

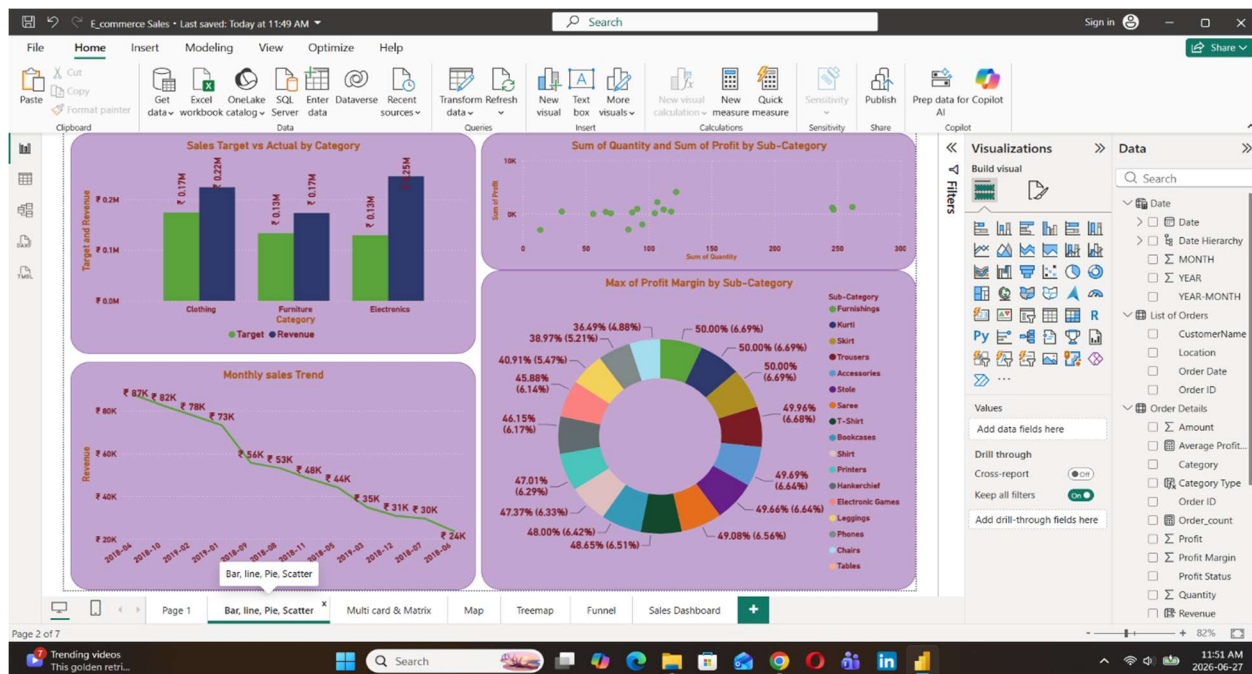
2. **Average Profit in Delhi-** Create a measure that calculates the average profit for orders placed in Delhi.

3. **Year-to-Date (YTD) Sales-** Create a measure that calculates total sales accumulated from the beginning of the year up to the current order date.



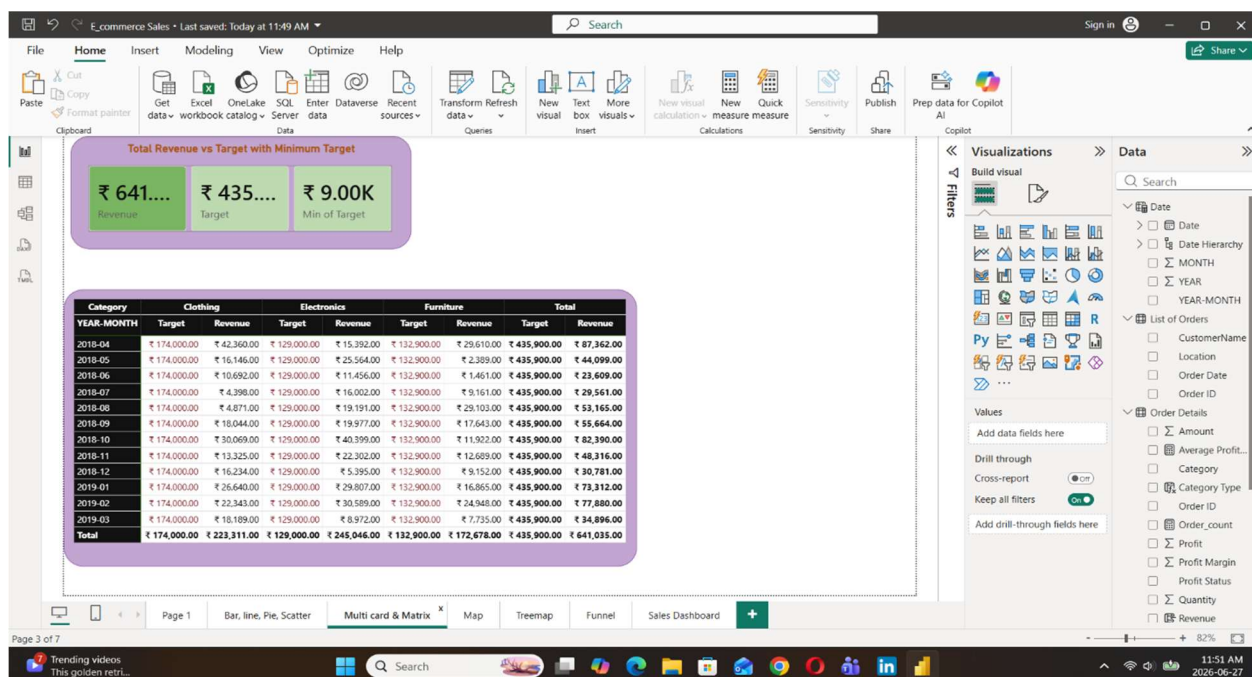
Data Visualization using Power BI:

- **Sales Target Achievement by Category:** Compare actual sales with sales targets by category using a clustered column chart.
- **Max Profit Margin by Sub-Category:** Analyze the maximum profit margin for each sub-category of products using a donut chart.
- **Monthly Sales Trend:** Show the trend of monthly sales over time using a line chart.
- **Comparison of Profit and Quantity by Sub-Category:** Compare the relationship between profit and quantity sold for different sub-categories using a scatter chart.

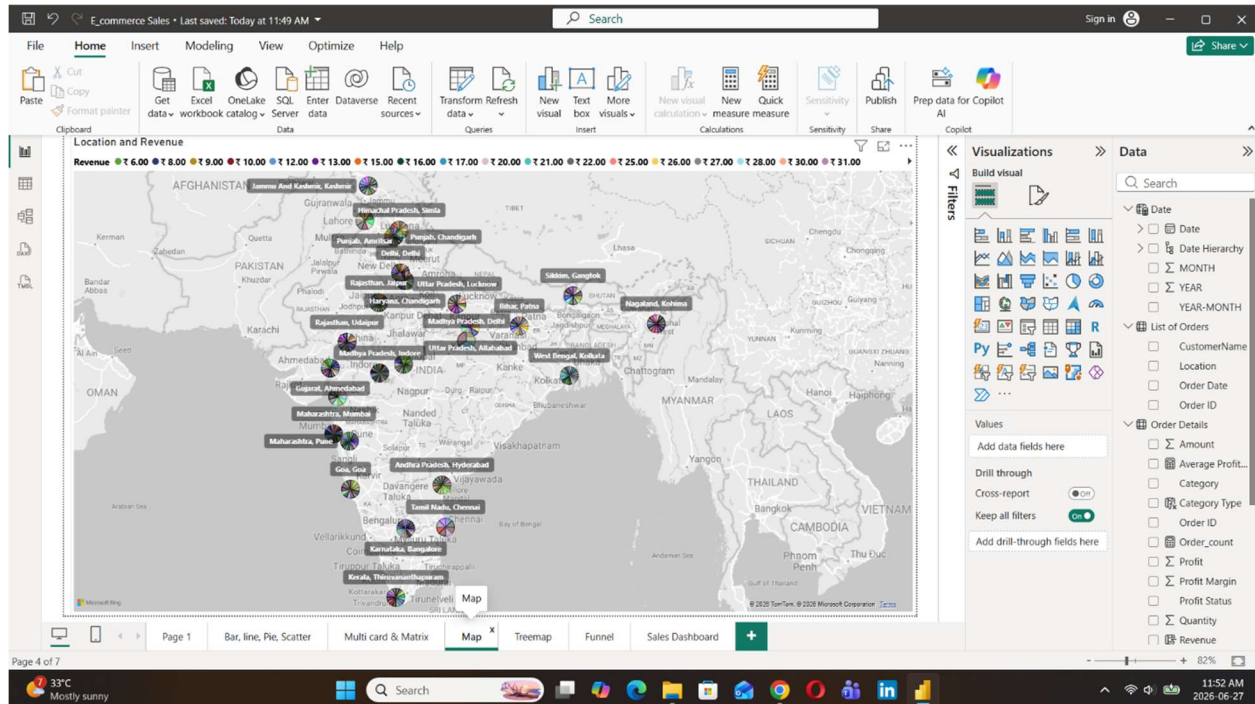


- **Comparison of Total Sales Amount and Target:** Create cards to succinctly display the total sales amount alongside the sales target for quick comparison and analysis. Also, create a multi-row card to display the minimum target for each segment.

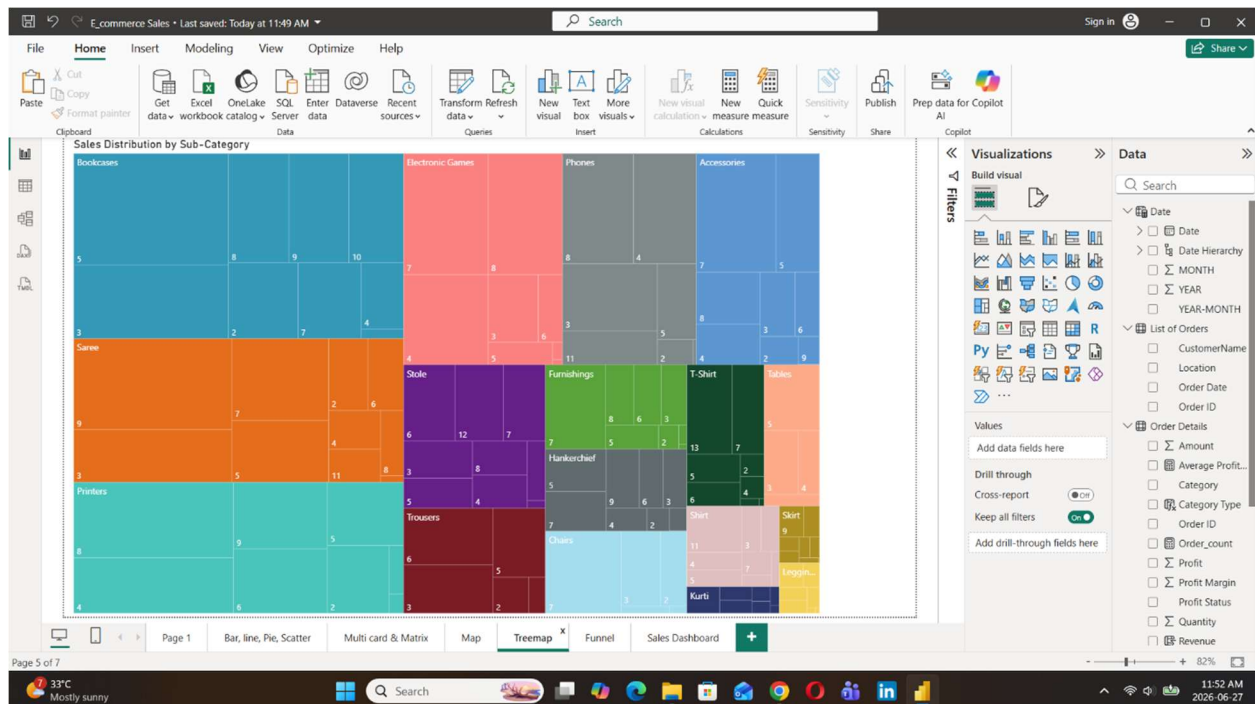
- **Sales Performance Matrix:** Build a matrix view to analyze how actual sales compare to sales targets across different categories and months.



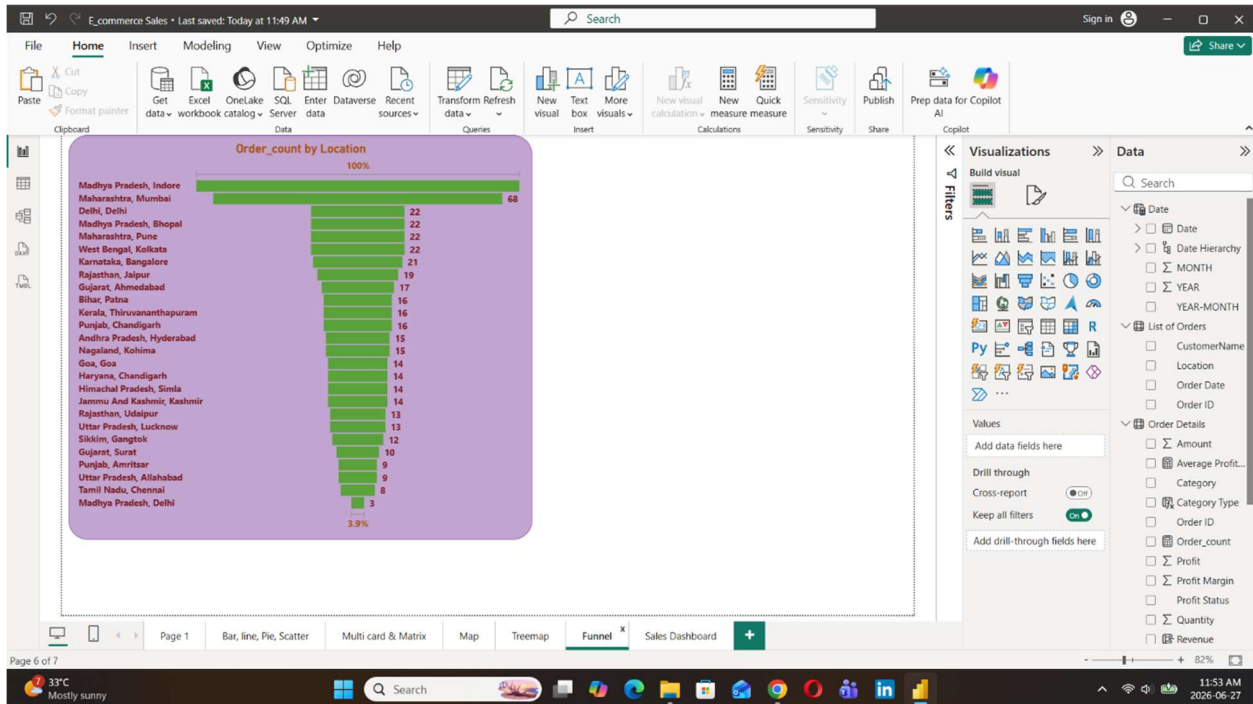
- **Geographic Sales Analysis:** Visualize total sales on a map by city to identify regional sales patterns.



- **Sales Distribution by Sub-Category:** Represent the sales distribution across different sub-categories using a Treemap.



- **Order Count Analysis by State:** Create a funnel chart to visualize the distribution of order counts across different states.



Sales Dashboard:

